NDSEG Research proposal Semi-final Draft

Granular materials packings have great potential to be used in shock and impact absorbing structures, due to their ability to absorb energy and continue to carry load after defects and even partial failures (1). However, dry packings are unwieldy to work with as they typically cannot be used as a building material on their own, without some external structure they are prone to losing their shape. My research goal is to engineer a new kind of programmable and re-programmable granular composite material that uses jamming to control mechanical properties. The composite I propose will be made from a honeycomb of cells with each cell filled with grains. The same volume of particles that can be crushed by the force from a hand, can support hundreds of newtons when jammed using just 20kPa confining pressure meaning. To program and reprogram the structural properties of the material the cell network will have pneumatic controls that enable the different combinations of cells to be jammed for different configurations. Also, after the composite has failed, it can be repaired by removing the load and rejamming, as the structures that fail will be the force chains within the cells. The goal set is to gain the greatest amount of control over the composite's mechanical properties by tuning the grains used, the honeycomb membranes, and the method by which the system is programmed.

Impact: This project stands to further jamming as a technique for programmable and self-healing materials for mesoscale applications. Practically the applications would be for those structures that are already designed to fail like impact absorbing structures with the benefit that they could be rebuilt from the same materials after failure.

Background: Prior work has shown the potential of granular media used as a structural material. Several projects have shown that with little to no external structure grains can be molded into load-bearing arches (2,3). However these structures are immobile, as they rely on gravity to maintain self confinement and therefore have limited flexibility outside of static structures. Also, while some have been shown to be tolerant to failure the entire structure must be rebuilt in order to return to where it was prior to failure (2, 3).

Other work has demonstrated the power of a confining membrane as a means to create a stable structure from loose materials (4). It has been shown that by jamming layers of spheres with a confining membrane, that structures as pedestrian bridges can be created (4). However, prior work has not explored the possibility to reconfigure the same material for multiple purposes and has not explored what properties could be obtained by optimizing the shapes and sizes of the particles used.

Jamming as a means to program a material has been demonstrated in the context of soft robotics, where segments of an actuator were jammed and unjammed to change the behavior of an actuator (5).

Research Plan: This project will be carried out in three sectioned goals.

Goal 1: The single cell system. Here the individual granular cell will be designed to have controllable mechanical properties. As has been demonstrated, different grains shapes can have very different responses to stress (6, 7, 8). In creating the cell the first goal is to accentuate the mechanical differences between the jammed and unjammed cell. Another important benchmark is the packing fraction on the jammed cell as this will set the amount of granular material needed. In order to find the particle shape that best fits this application using covariance matrix adaptation evolutionary strategy (CMA-ES) to optimize the shape for a specific mechanical property (9, 10). In addition to particle shape, the preparation of the cell will also be explored as an avenue to control the mechanical properties. Preliminary results have shown that it is possible to induce alignment with a chosen axis by vibrating a packing with a lab mixer. This project would explore the possibility of using this alignment control to select one axis to have different properties than the other two when a cell is jammed.

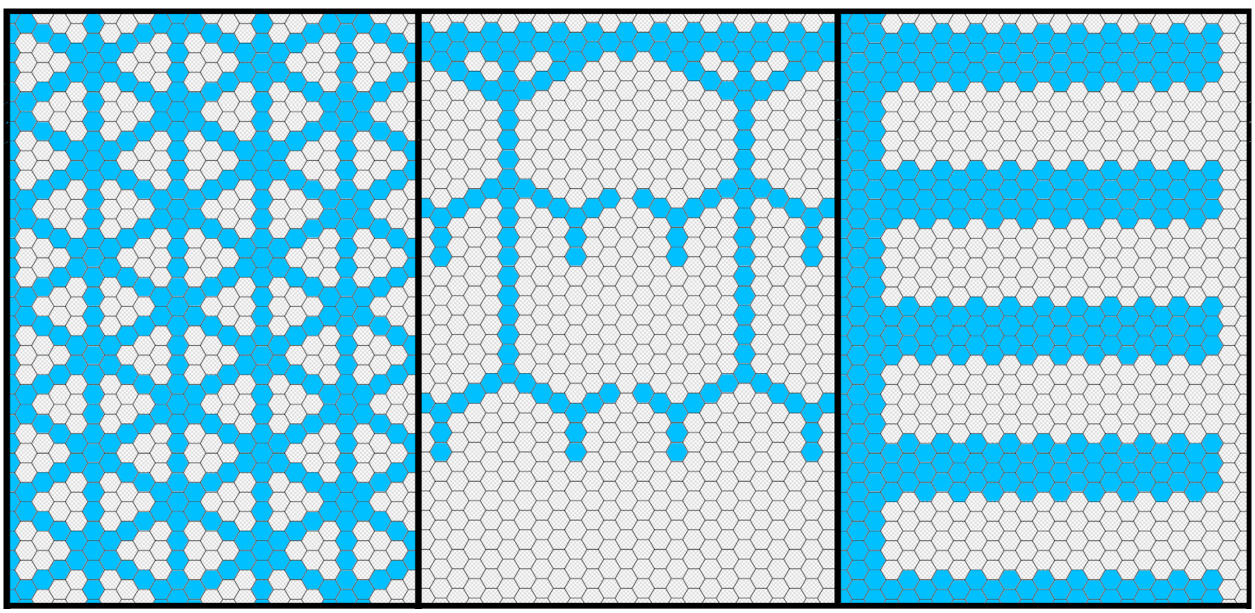
A milestone would be if we are able to create a large difference in shear strength between jammed and unjammed or select directionality of the maximum shear strength, either of which would allow these cells to be used as programmable building blocks. Optimization would be done using an N spheres model to represent particle shapes and a DEM simulation to test mechanical properties. Before proceeding to Goal 2, the simulated results will be validated using real world particles printed on an Objet Connex350 printer. If unable to produce new particles that have record differences in jammed and unjammed states the project will proceed with particles previously known to have large differences. Once finalized the single cell system will be characterized using an Instron for use in later modeling.

Goal 2: The Honeycomb Here the network that houses the cells will be developed. Once the cells are created the network that separates the cells must be developed. The goal here is to find a material that is airtight and shows high-tensile strength. Several materials have been used to enclose jammed grains, and to form evacuated networks (4). The search will begin with commercially available PDMS, Dragon Skin, and ETFE foil.

The control system must also be developed. Using simulation, this time to model the honeycomb with a simplified model for each cell, configurations of the network will be optimized for the useful and different mechanical functions, such as impact dissipation, strength under shear, normal and bending stress, and predictable failure. With these configurations found the controls of the network can be developed. It would not be feasible to have each cell controlled individually as that would require N^3 pneumatic control lines, so it is vital to reduce the number of isolated sections as much as possible. Knowledge of the configurations will allow cells to be grouped into blocks or sections with a single control while still enabling a wide variety of different configurations.

Goal 3: Characterization and scalability The last step will be to characterize the system. Creating the composite using the methods developed for goals one and two and test it in various configurations. It will be tested for ease of programming and how many cycles it can be configured.

Figure:



Cross section of what various configurations might look like for isotropic strength, impact dissipation, and bending. Blue indicates jammed cells, grey unjammed.

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